

The original integral signed order and its exact normalization

Exterior channel E



$$C_p = \text{diag}(V, \text{diag}(p^{m_i})V)$$

$$b = (1, 1, 0, 0), \quad m = (0, 0, 0, 0), \quad \varepsilon = (1, 1, 0, 0)$$

$$\text{Smith}(C_p) = (0, 0, 0, 0, 0, 0, 1, 1)$$

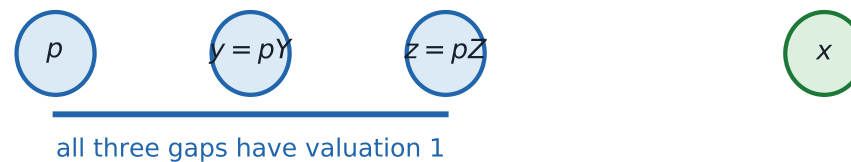
$$\tilde{\mathcal{E}}_p / \mathcal{E}_p \simeq (\mathbb{Z}_p / p)^2$$

$$[\tilde{\mathcal{E}}_p : \mathcal{E}_p] = p^2$$

$$V_{\{p, z\}} = \mathbf{1} \oplus \chi_{pD_E}^{\oplus 2}, \quad a = 2$$

Integral defect and ramification are computed from the same factors;
neither invariant is substituted for the other.

Middle channel M



$$C_p = \text{diag}(V, \text{diag}(p^{m_i})V)$$

$$b = (2, 2, 2, 0), \quad m = (1, 1, 1, 0), \quad \varepsilon = (0, 0, 0, 0)$$

$$\text{Smith}(C_p) = (0, 0, 0, 1, 1, 2, 2, 3)$$

$$\tilde{\mathcal{E}}_p / \mathcal{E}_p \simeq (\mathbb{Z}_p / p)^2 \oplus (\mathbb{Z}_p / p^2)^2 \oplus \mathbb{Z}_p / p^3$$

$$[\tilde{\mathcal{E}}_p : \mathcal{E}_p] = p^9$$

$$\det(1 - T\text{Frob} \mid V) = (1 - T)^5 \text{ or } (1 - T)^3(1 + T)^2$$

Integral defect and ramification are computed from the same factors;
neither invariant is substituted for the other.